Chapter 2 - The Real Number System

Problem Set # 3

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1. Mark each statement True or False. Justify each answer.

   (a) If a nonempty subset of $\mathbb{R}$ has an upper bound, then it has a least upper bound.

   (b) If a nonempty subset of $\mathbb{R}$ has an infimum, then it is bounded.

   (c) Every nonempty bounded subset of $\mathbb{R}$ has a maximum and a minimum.

   (d) If m is an upper bound of S and $m' < m$, then m' is not an upper bound of S.

   (e) If $m = \inf S$ and $m' < m$, then m' is a lower bound of S.

   (f) For each real number x and each $\varepsilon > 0$, there exists $n \in \mathbb{Z}^+$ such that $n\varepsilon > x$.

2. Mark each statement True or False. Justify each answer.

   (a) Every nonempty subset of $\mathbb{Z}^+$ contains a minimum.

   (b) Every nonempty subset of $\mathbb{Z}^+$ contains a maximum.

   (c) If x and y are irrational, then xy is irrational.

   (d) Between any two unequal rational numbers there is an irrational number.

   (e) Between any two unequal irrational numbers there is a rational number.

   (f) The rational and irrational numbers alternate, one then the other.

3. For each subset of $\mathbb{R}$, give its supremum and its maximum, if they exist. Otherwise, write "none."

   (a) $\{1, 3\}$

   (b) $\{\pi, 3\}$

   (c) $[0, 4]$

   (d) $(0, 4)$

   (e) $\left\{\frac{1}{2n} : n \in \mathbb{Z}^+\right\}$

   (f) $\left\{1 - \frac{1}{n} : n \in \mathbb{Z}^+\right\}$

   (g) $\left\{\frac{n}{n+1} : n \in \mathbb{Z}^+\right\}$

   (h) $\left\{(-1)^n \left(1 + \frac{1}{n}\right) : n \in \mathbb{Z}^+\right\}$

   (i) $\left\{\frac{n+(-1)^n}{n} : n \in \mathbb{Z}^+\right\}$

   (j) $(-\infty, 4)$

   (k) $\bigcap_{n=1}^{\infty} \left(1 - \frac{1}{n}, 1 + \frac{1}{n}\right)$

   (l) $\bigcup_{n=1}^{\infty} \left[n, 2 - \frac{1}{n}\right)$

   (m) $\{r \in \mathbb{Q} : r < 5\}$

   (n) $\{r \in \mathbb{Q} : r \leq 5\}$

4. Repeat Exercise 3 for the infimum and the minimum of each set.

5. Let S be a nonempty bounded subset of $\mathbb{R}$ and let $m = \sup S$. Prove that $m \in S$ iff $m = \max S$.

6. (a) Let S be a nonempty bounded subset of $\mathbb{R}$. Prove that $\sup S$ is unique.

   (b) Suppose that m and n are both maxima of a set S. Prove that $m = n$.

7. Let S be a nonempty bounded subset of $\mathbb{R}$ and let $k \in \mathbb{R}$. Define $kS = \{ks : s \in S\}$. Prove the following:

   (a) If $k \geq 0$, then $\sup(kS) = k \cdot \sup S$ and $\inf(kS) = k \cdot \inf S$.

   (b) If $k < 0$, then $\sup(kS) = k \cdot \inf S$ and $\inf(kS) = k \cdot \sup S$.

8. Let S and T be nonempty bounded subsets of $\mathbb{R}$ with $S \subseteq T$. Prove that

$$\inf T \leq \inf S \leq \sup S \leq \sup T.$$

9. (a) Prove: If $y > 0$, then there exists $n \in \mathbb{Z}^+$ such that $n - 1 \leq y < n$.

   (b) Prove that the n in part (a) is unique.

10. (a) Prove: If x and y are real numbers with $x < y$, then there are infinitely many rational numbers in the interval $[x, y]$.

    (b) Repeat part (a) for irrational numbers.

11. Let y be a positive real number. Prove that for every $n \in \mathbb{Z}^+$ there exists a unique positive real number x such that $x^n = y$.

12. Let D be a nonempty set and suppose that $f : D \to \mathbb{R}$ and $g : D \to \mathbb{R}$. Define the function $f + g : D \to \mathbb{R}$ by $(f + g)(x) = f(x) + g(x)$.

    (a) If $f(D)$ and $g(D)$ are bounded above, then prove that $(f + g)(D)$ is bounded above and

$$\sup\left((f + g)(D)\right) \leq \sup f(D) + \sup g(D).$$

    (b) Find an example to show that a strict inequality in part (a) may occur.

    (c) State and prove the analog of part (a) for infima.

13. Let $x \in \mathbb{R}$. Prove that $x = \sup\{q \in \mathbb{Q} : q < x\}$.

14. Let a/b be a fraction in lowest terms with $0 < a/b < 1$.

    (a) Prove that there exists $n \in \mathbb{Z}^+$ such that

    $$\frac{1}{n+1} < \frac{a}{b} < \frac{1}{n}.$$

    (b) If n is chosen as in part (a), then prove that $a/b - 1/n$ is a fraction that in lowest terms has a numerator less than a.

    (c) Use part (b) and the principle of strong induction to prove that a/b can be written as a finite sum of distinct unit fractions:

    $$\frac{a}{b} = \frac{1}{n_1} + \cdots + \frac{1}{n_k}$$

    where $n_1, \ldots, n_k \in \mathbb{Z}^+$. (This is related to the Egyptian system of arithmetic.)

15. Prove Euclid's division algorithm: If a and b are natural numbers, then there exist unique numbers q and r, each of which is either 0 or a positive integer, such that

    $$r < a \quad \text{and} \quad b = qa + r.$$

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"Great works are performed not by strength, but by perseverance." - Samuel Johnson -